

Homomorphic Encryption

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1. Introduction:

The issue of data privacy and security has long been problematic in the field of computer science. Commonly, practices such as legacy encryption have been used as a method of cloud cybersecurity. This method of encryption relies on a shared key which only the parties involved can use to decode the encrypted pieces of data. However, several problems arise when utilizing this type of encryption. Firstly, the data can be accessed not only by the parties involved, but also by the third party service providers who can view and steal this information. Therefore, any sensitive data can be accessed by these servers and can continue to be viewed for long periods of time after the customer discontinues the use of these servers (Acar et al., 2017). The widespread cultural dependence on technology will inevitably lead to a higher demand for appropriate measures to be taken in order to guarantee security and privacy of each user's data. This is where the revolutionary technology of Homomorphic Encryption comes into play. This research essay will delve into the specifics of this new technology and discuss several possible solutions for how to implement this form of encryption effectively in practical applications.

With normal encryption, a piece of data (known as the plaintext) is altered using an algorithm into an obfuscated form (called the ciphertext). The algorithm used determines how secure the encryption is. Algorithms that are infeasible to brute-force computationally, like the factors of extremely large integers, are ideal for this purpose. However, with normal forms of encryption the data cannot be changed while it is encrypted. Homomorphic Encryption improves the security of data stored online because the data does not need to be decrypted for certain operations to be performed on it. This reduces the number of opportunities that a bad actor could use to steal the data while it is unencrypted. As an example, a bank could use Homomorphic

Encryption to store financial information for its clients, allowing certain financial transactions to be made without decryption being necessary.

2. Body:

For this literature review, five different articles pertaining to Homomorphic Encryption were chosen. Two of these articles were about specific limitations that Homomorphic Encryption has, and possible solutions to these limitations. Two were about recent advances in Homomorphic Encryption in general, and about the difference between full and partial Homomorphic Encryption. The final article was about Homomorphic Encryption as it relates to quantum cryptography. The paragraphs about these articles were grouped together accordingly. The research in this essay will attempt to answer the question: What practical steps can be taken to implement the new technology of Homomorphic Encryption in order to revolutionize the fields of cybersecurity and data management?

There is a distinction to be made between fully Homomorphic Encryption and partially Homomorphic Encryption. A fully Homomorphic Encryption scheme “allows any computable function to perform on the encrypted data” (Acar et al., 2017). In contrast, a partially Homomorphic Encryption scheme only allows certain operations to be performed. Although Homomorphic Encryption has been a concept for over 30 years, the first fully Homomorphic Encryption scheme was only developed in 2009 (Acar et al., 2017). Although this initial implementation had many flaws, it represents an important discovery in the field of cryptography.

Since 2009, Homomorphic Encryption has been improving at a rapid pace. The time taken to perform multiplication on Homomorphically Encrypted ciphertexts has “dropped from

30 minutes to less than 20 milliseconds” in the last decade (Viand et al., 2021). This is an improvement of almost five orders of magnitude. A real-world example of an operation in which Homomorphic Encryption has been implemented is in the field of medicine. It was used in allowing for “privacy-preserving genome analysis” which ran through and encrypted enormous quantities of data in a timely manner, demonstrating its ever increasing practicality (Viand et al., 2021). It has also been used in small-scale mobile applications, such as fitness apps, allowing for increased privacy through the back end encryption technology without affecting the user experience (Viand et al., 2021). However, there are also some challenges associated with fully Homomorphic Encryption. For one thing, Homomorphic Encryption schemes are by definition incompatible with “data-dependent branching,” such as the if-else statements common in virtually all modern programming languages (Viand et al., 2021). And although Homomorphic Encryption has become much faster since 2009, operations on the ciphertext are still about seven orders of magnitude slower than they would be on the plaintext (Viand et al., 2021).

Partially Homomorphic Encryption also has some limitations, however. Each operation that is performed on encrypted data causes some noise to accumulate, and once enough noise is accumulated, the data can no longer be decrypted (Reis et al., 2020). In such a case, the original data can only be retrieved from a backup, if a backup even exists. Different operations cause different amounts of noise to accumulate. Multiplication, for example, causes much more noise than addition does (Reis et al., 2020). This noise is only a problem for certain kinds of Homomorphic Encryption. The encrypted data is also much, much larger, which means the computations on encrypted data will often be far slower than they would be if the data wasn’t encrypted. One possible solution to this problem is the technique of computing-in-memory. Computing-in-memory is a way of designing computers, where the circuits used for arithmetic

and logical operations are located in the memory chip itself (Reis et al., 2020). This allows for far faster computation than would otherwise be possible. This approach is not without flaws. Computing-in-memory has not yet been implemented for certain arithmetic operations, such as multiplication of large numbers and modulus by powers of two (Reis et al., 2020). Nonetheless, Homomorphic Encryption in combination with computing-in-memory shows a great deal of potential for use in cybersecurity.

Another limitation of Homomorphic encryption is the selection of parameters to use. The choice of parameters has a massive impact on the security, precision, and performance of a given Homomorphic Encryption system (Cabrero-Holgueras et al., 2023). The efficacy of the parameters themselves also depends on the design specifications of the computer it is running on. Because of the degree of complexity involved, it is very difficult to determine an optimal set of parameters by hand. Cabrero-Holgueras et al. (2023), propose a method for optimal parameters to be determined mathematically, using a combination of linear programming and fuzzy logic. Fuzzy logic is a form of boolean algebra where truth values may be a value between zero and one, rather than only zero or one. By using fuzzy logic, this method allows precision, security, and performance to be prioritized and constrained in a rigorous, mathematical way (Cabrero-Holgueras et al., 2023). Linear programming is a way of solving optimization problems. The outputs of the fuzzy logic are combined with detailed information about the architecture of the computer system that the Homomorphic Encryption scheme will be running on, and linear programming is then used on this combined information to find an optimal set of parameters. This research by Cabrero-Holgueras et al. helps to make Homomorphic Encryption more accessible and easier to implement.

In recent years, the possibility of conventional encryption schemes being rendered insecure by quantum computing has become more and more likely. However, some encryption algorithms have been developed which take advantage of quantum computing for increased efficiency and/or security. Can Homomorphic Encryption likewise benefit from quantum computing? According to Li et al. (2014), there are some limitations on quantum Homomorphic Encryption. More precisely, “Current schemes only achieve approximate perfect security, however, and result in encodings which grow exponentially as they approach perfect security” (Li et al., 2014). In other words, it is impossible to create a quantum Homomorphic Encryption scheme that is both efficient and perfectly secure. However, this limitation only applies to ‘full’ Homomorphic Encryption schemes, where any boolean or arithmetic operation can be performed on the ciphertext (Li et al., 2014). In practice, this is not necessary for many real-world applications, so quantum Homomorphic Encryption may still have many useful applications.

3. Summary:

In summary, finding an effective way to implement Homomorphic Encryption is a critical step in the future of cybersecurity. The nuances of its algorithm design and parameter selection make it highly complex and difficult to efficiently incorporate. (Cabrero-Holgueras et al., 2023) Despite these various limitations, how can this encryption tool be used to create a perfect security system for these kinds of data transmissions? The general consensus in order to “overcome the challenges of HE has come in the form of compilers, frameworks, and APIs.” (Viand et al., 2021, pg. 1) Solutions such as Computing-in-Memory (CiM), the combination of fuzzy logic and linear programming, and advancements in the field of quantum computing have greatly increased the practicality and potential of this technology. (Reis et al. 2020) This style of

encryption is by no means perfected. However, the creative solutions to effectively performing these complex algorithms on encrypted data combined with the drastic increase in bootstrapping speed as a result of these methods promises for rapid strides in the implementation of Homomorphic Encryption. Gartner, a popular technology review, predicts that “by 2025, at least 20% of companies will have a budget for projects that include fully homomorphic encryption.” (Driver, 2020) By this metric, it is undeniable that the future is bright for this new technology of Homomorphic Encryption and its implementation will soon be rapidly spread within the world of cybersecurity.

4. References:

- Acar, A., Aksu, H., Uluagac, A. S., & Conti, M. (2017, October 6). *A survey on homomorphic encryption schemes: Theory and implementation*. arXiv.org. <https://arxiv.org/abs/1704.03578>
- Cabrero-Holgueras, J., & Pastrana, S. (2023). Towards automated homomorphic encryption parameter selection with fuzzy logic and linear programming. *Expert Systems with Applications*, 229, 120460. <https://doi.org/10.1016/j.eswa.2023.120460>
- Reis, D., Takeshita, J., Jung, T., Niemier, M., & Hu, X. S. (2020). Computing-in-memory for performance and energy-efficient homomorphic encryption. *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, 28(11), 2300–2313. <https://doi.org/10.1109/tvlsi.2020.3017595>
- Yu, L., Pérez-Delgado, C., & Fitzsimons, J. (2014). Limitations on information-theoretically-secure quantum homomorphic encryption. *Physical Review A*, 90(5). <https://doi.org/10.1103/physreva.90.050303>
- Viand, A., Jattke, P., & Hithnawi, A. (2021). *SOK: Fully homomorphic encryption compilers* | *IEEE conference ... SoK: Fully Homomorphic Encryption Compilers*. <https://ieeexplore.ieee.org/document/9519484/>
- Driver, M., “Emerging technologies: *Homomorphic encryption for data sharing with privacy*,” Gartner, Inc, Tech. Rep. 23 April, 2020.